

NORTHERN UNIVERSITY BANGLADESH

LAB REPORT - 01

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Project Name : Smart Inventory & Supply Chain Management System

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Table of Contents

Abstract.....	2
Acknowledgment.....	2
List of Figures & Table	3
Introduction.....	4
Literature Review	4
Methodology / Implementation.....	4
Innovation & Uniqueness	5
Result & Discussion	6
Application & Future Scope	6
Conclusion	7
References / Bibliography	7

Abstract

The "Smart Inventory & Supply Chain Management System" is designed to optimize the flow of goods, materials, and data across an organization. The core goal is to manage inventory levels efficiently, streamline supplier coordination, and ensure real-time visibility into stock and order movements. The project primarily focuses on designing and developing a well-structured relational database, which will serve as the backbone of the entire system.

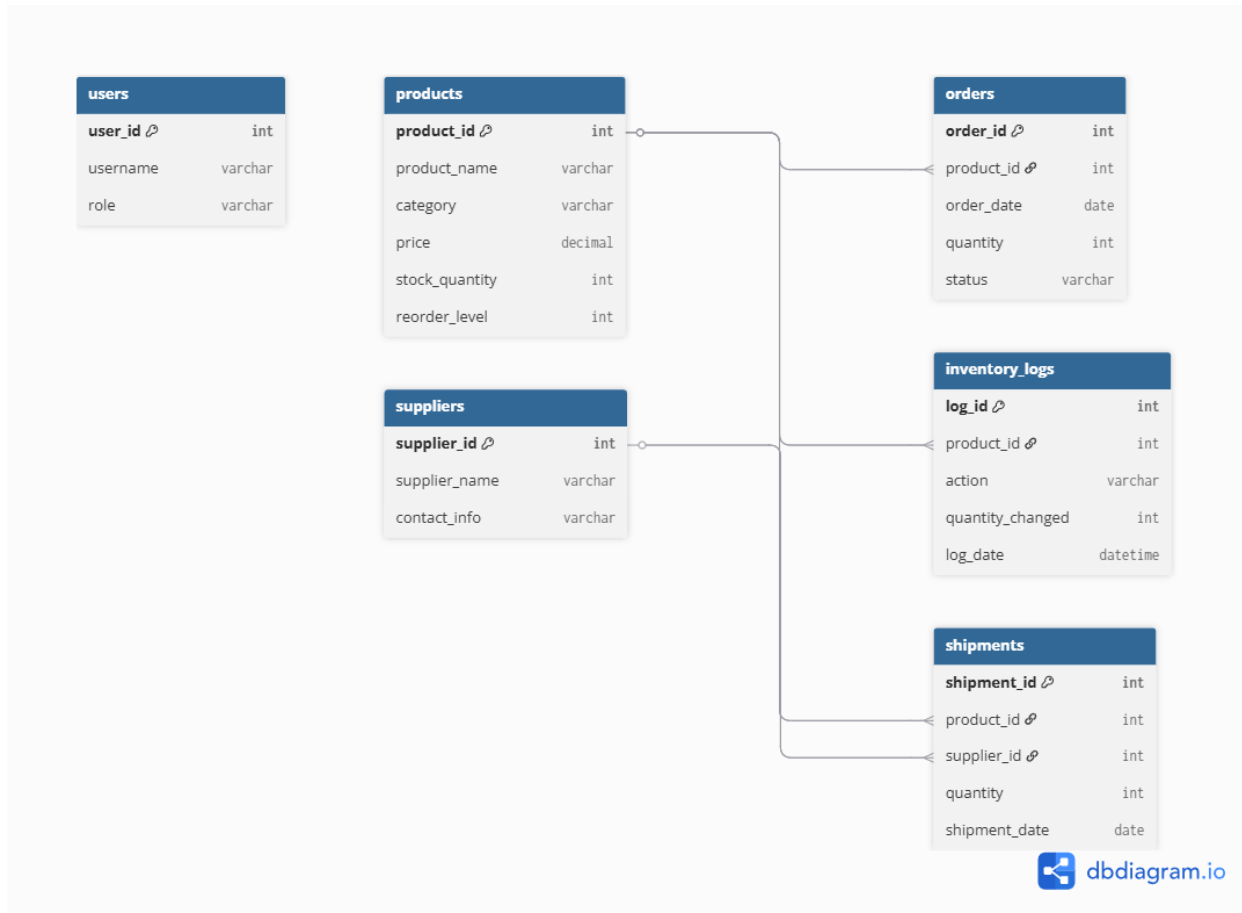
Utilizing structured query language (SQL) and procedural extensions, the project enables efficient data handling, transaction management, and advanced querying. Additionally, a user interface is optionally implemented to visually present and simulate inventory operations. This project ensures accurate tracking, timely reordering, and better coordination between suppliers and warehouse units. The outcome will demonstrate how database systems can be used to solve real-world logistics challenges and enhance business productivity

Acknowledgment

We express our sincere gratitude to our course instructor, Tasfia Tabassum Faija, for providing us with the opportunity to undertake this project and for the valuable guidance throughout the development. We would also like to thank our peers and academic coordinators for their constructive feedback and encouragement. Lastly, we appreciate the support from our institution and lab facility that enabled us to carry out our work effectively.

List of Figures & Table

- Figure 1: Entity-Relationship Diagram (ERD)



- **Table 1:** products Table Schema
- **Table 2:** suppliers Table Schema
- **Table 3:** orders Table Schema
- **Table 4:** shipments Table Schema
- **Table 5:** inventory logs Table Schema
- **Table 6:** users Table Schema
- **Table 7:** Sample JOIN Query Result (Products vs Suppliers)

Introduction

Inventory and supply chain management are critical components of any business operation dealing with physical goods. Inefficiencies in these systems can lead to overstocking, stock outs, or delays — all of which impact customer satisfaction and profitability. The objective of this project is to design and simulate a Smart Inventory & Supply Chain Management System using relational database design principles and advanced SQL features. By structuring the data and logic effectively, businesses can automate reordering, monitor supplier performance, and manage stock levels proactively.

Literature Review

Traditional inventory systems often rely on static spreadsheets or isolated software without dynamic tracking capabilities. Recent advancements have introduced ERP and cloud-based inventory tools that allow real-time data access and integration with suppliers. Our project differentiates itself by emphasizing a backend-driven architecture with enhanced procedural logic using SQL triggers and procedures. Unlike traditional systems, it includes capabilities like automatic reorder triggers, supplier-to-product mapping, and transaction logging, thus providing a smart and scalable inventory management experience.

Methodology / Implementation

Tools & Technologies:

- Database: Relational SQL-based database (e.g., Oracle, MySQL, or PostgreSQL)
- Query Language: SQL + Procedural extensions (like PL/SQL)
- Frontend (Optional): React.js for UI simulation

Process Overview:

- **Step 1:** Requirements gathering and use-case analysis
- **Step 2:** Designing ER diagram and schema structure
- **Step 3:** Creating tables and inserting sample data using `schema.sql`
- **Step 4:** Writing queries, triggers, procedures, and reports in `logic_queries.sql`
- **Step 5:** (Optional) Developing a UI to demonstrate core features like viewing products, orders, and supplier data.

Innovation & Uniqueness

This project introduces several smart features typically absent in beginner-level inventory systems:

- Dynamic stock level monitoring and reorder logic through triggers
- Supplier-product mapping using JOIN queries for sourcing analysis
- Use of stored procedures to encapsulate order processing logic
- Optional React-based UI for live demonstration of backend logic

In contrast to static inventory spreadsheets or basic CRUD apps, this system actively monitors, reacts, and reports on stock and order conditions in real time.

Result & Discussion

The developed database schema successfully captures inventory, suppliers, shipments, orders, and related data through normalized tables. Complex queries using JOIN operations provide integrated views across multiple entities, such as order fulfillment efficiency or low-stock alerts. Trigger-based stock updates were tested for various order scenarios, and stored procedures were written to streamline reorder logic. Challenges included learning procedural SQL and ensuring relational integrity, which were overcome through incremental testing and ER-based planning.

Application & Future Scope

Real-world use cases include:

- Retail store inventory systems
- Warehouse management
- Manufacturing resource planning (MRP)
- E-commerce backend logistics

Future enhancements may involve:

- Live backend integration via Node.js/Express
- User roles (admin/supplier views)
- Data analytics dashboard for inventory trends
- Real-time email/SMS notifications for stock outs

Conclusion

This project demonstrates how structured database systems can greatly enhance the efficiency and intelligence of inventory and supply chain operations. With normalized tables, procedural logic, and query optimization, we built a smart solution that addresses real-world supply chain needs. While frontend development was optional, its inclusion further illustrates how database systems serve as the backbone of full application workflows. The learning and implementation have equipped us with practical skills in SQL-based system design and backend logic structuring

References / Bibliography

- Oracle Documentation: <https://docs.oracle.com/en/database/>
- MySQL Reference Manual: <https://dev.mysql.com/doc/>
- Database System Concepts – Abraham Silberschatz
- TutorialsPoint SQL & PL/SQL Tutorials
- W3Schools SQL Documentation